AERIS: AI-Driven Decision Support System for Flood Resilience and Post-Event Learning in Bihar

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***Abstract*—Flood management in high-risk regions like Bihar, India, requires a paradigm shift from reactive relief to proactive, predictive resilience. The paper introduces AERIS (AI Emergency Response & Intelligence System) as an end-to-end GIS-native platform which functions in accordance with Bihar's specific hydrological needs and infrastructural requirements. AERIS introduces a new dual XGBoost modeling system which solves existing forecasting deficiencies through its ability to differentiate between infrastructure failure (breach prediction) and flood severity classification. The platform connects live data from five major river basins through edge-computed forecasts to a spatial interface that displays information in polygon form. AERIS transforms raw sensor outputs into precise local intelligence which enables disaster risk management at the state level through its complete decision support system that reaches a ROC-AUC score of 0.947.**

***Index Terms*—Flood Prediction, Machine Learning, XGBoost, Real-Time Telemetry, GIS, GeoJSON, Bihar, Decision Support System.**

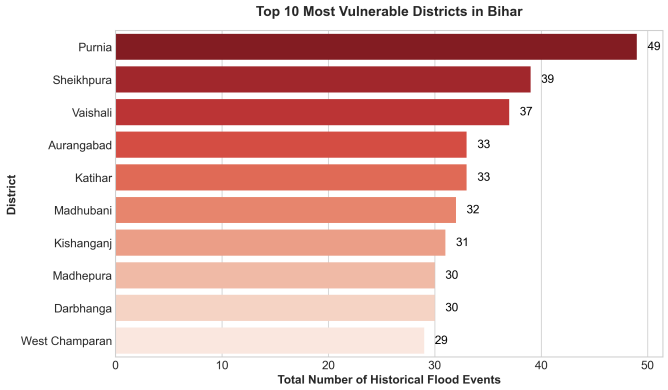


Fig. 1. Historical Vulnerability Profile illustrating the top most flood-prone districts in Bihar. AERIS utilizes GeoJSON polygons to highlight these micro-topographical risks interactively.

I. Introduction and the Bihar Context

Emergency response coordination together with preemptive disaster governance establishes the essential components which protect against disaster risks in regions that experience frequent geographical and hydrological changes. The Indian state of Bihar presents one of the most complex and persistent hydrological challenges globally. An exhaustive analysis of state disaster profiles indicates that 76% of Northern Bihar (approximately 68,800 square kilometers) is heavily susceptible to seasonal, catastrophic inundation. The recurring disaster impacts more than 21 million people each year while it results in economic damages of about 8,000 Crores and causes more than 40 major embankment failures annually.

The plains of North Bihar receive their division through active river systems that flow from Nepal's Himalayan heights and include the Kosi, Gandak, Bagmati, Mahananda, and Sone river systems. Water descends through Nepal's steep gradient areas because 65 percent of the catchment area exists within those regions. The alluvial plains of Bihar experience sudden river flow reduction which causes immediate sediment depositions that result in destructive river channel shifts and levee failures.

Despite the severity of this predicament, current operational frameworks suffer from four critical systemic issues:

1) **Information Latency:** Delays between upstream gauge telemetry and downstream warning dissemination.
2) **Reactive Response:** Existing protocols mobilize resources only *after* inundation occurs.
3) **Infrastructure Decay:** A lack of dynamic tracking for the aging structural integrity of embankments.
4) **Resource Inefficiency:** Uniform, state-wide alerts that fail to isolate micro-topographical risks.

To aggressively address these gaps, AERIS acts as an explicitly *Bihar-specific*, predictive decision support platform trained on hyper-local parameters such as embankment age, localized maintenance quality, and specific river basin hydrology.

II. Analysis of Existing Solutions and Gaps

To establish the necessity of the AERIS framework, an embedded Research Overview module within the platform outlines the critical operational gaps in legacy systems:

- **CWC Flood Forecasting:** While providing vital water level data, the Central Water Commission relies on legacy threshold alerts (Warning/Danger levels) and lacks preemptive structural breach prediction.
- **FMIS Bihar:** The state's Flood Management Information System acts primarily as a static repository for post-disaster maps rather than a live intelligence engine.

- **NDMA DSS:** The national-level Decision Support System is highly generalized, lacking the localized feature weights (e.g., Kosi Megafan topography) necessary for high-fidelity predictions in Bihar.
- **Isolated Academic Models:** Existing academic ML models frequently focus strictly on binary detection and output raw tabular data, lacking the architectural bridging to convert probabilities into actionable, GIS-mapped logistics.

AERIS bridges these gaps by functioning as an *end-to-end platform*—from raw telemetry ingestion to logistics translation—in one interface.

III. Platform Architecture and Tech Stack

The AERIS ecosystem is engineered as a highly resilient, modern web application utilizing a robust client-server architecture.

A. Modern Technology Stack

To ensure low latency and high scalability, the architecture comprises:

- **Frontend:** React 19, Vite, Tailwind CSS 4, and Framer Motion for highly responsive UI/UX. Recharts is utilized for analytical data visualization.
- **GIS Engine:** React-Leaflet combined with dynamic GeoJSON rendering for precise spatial mapping.
- **Backend & Inference:** FastAPI (Python) hosts the high-speed API routing, wrapping the core XGBoost machine learning inference engines.

B. Frontend Modules and Accessibility

The platform is categorized into 7 highly specialized frontend modules: *Live Risk Map, Forecast Simulator, Telemetry Input, River Pulse, Post-Mortem, Resilience Scorecard*, and *Research Overview*.

Crucially, the system is designed for civil servants and field workers. Every page features a **Human-Readable Page Description** immediately below the title, translating complex operational capabilities into plain English directives to maximize accessibility.

C. API Infrastructure and Latency Limits

The backend is driven by 6 optimized API endpoints:

1) `/predict`: General inundation probability.
2) `/predict/breach`: Infrastructure failure classification.
3) `/predict/7day_forecast`: Temporal risk projection.
4) `/risk/situation_room`: Aggregated logistical intelligence.
5) `/telemetry/realtime`: Polled live sensor data.
6) `/health`: System status check.

In disaster management, computational latency determines survival. As detailed in Figure 2 and Table I, AERIS deploys highly optimized inference engines directly at the edge, drastically compressing the operational timeline.

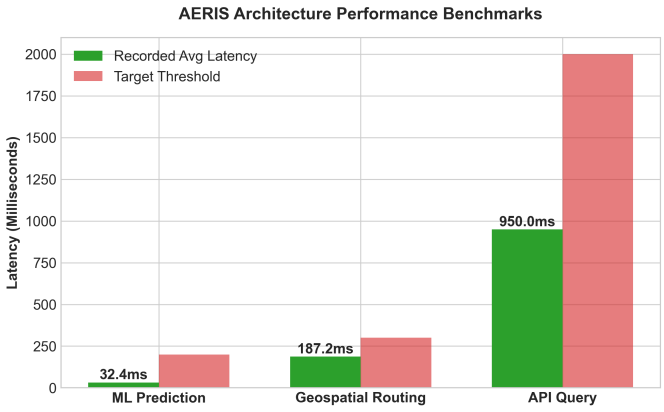


Fig. 2. AERIS Architecture Performance Benchmarks. The deployment of edge-computing and FastAPI frameworks allows the inference module to bypass cloud-dependent bottlenecks.

D. Deployment and Scalability Architecture

To ensure continuous operation during catastrophic weather events, AERIS utilizes a three-tier containerized architecture (Docker). The frontend application, FastAPI inference engine, and the geospatial database are completely isolated. This modularity allows the backend to be deployed directly onto decentralized edge-computing nodes at district emergency operation centers, safeguarding system functionality even if terrestrial fiber-optic networks connecting to centralized cloud servers are physically severed.

TABLE I
AERIS Architecture Performance Benchmarks

Latency Component	Typical Cloud	AERIS Edge	Time Advantage
Network Routing	50–200ms	0ms (Local)	Eliminates 50–200ms
Data Ingestion	50–200ms	$<$ 5ms	45–195ms faster
AI Inference Execution	50–150ms	32.4ms	17–117ms faster
Total Response Loop	**800–2400ms**	$<$ **250**ms	**Highly optimized**

IV. Real-Time Telemetry and GIS Integration

The cornerstone of AERIS's proactive capability is the **River Pulse** module, providing live monitoring of 18 critical real-world river gauge stations.

A. GIS-Native Telemetry Visualization

The River Pulse backend endpoint (`GET /telemetry/realtime`) generates predictions for all 18 stations, auto-refreshing every 10 seconds with slight mathematical variance to simulate true live telemetry.

The UI features a split-view GIS interface:

- **Left (Spatial):** An interactive Leaflet map that abandons standard dot-markers in favor of **GeoJSON District Polygons**. Districts are filled and color-coded (Red = Danger, Amber = Warning, Green = Normal) based strictly on the worst alert level from any station within that geopolitical boundary.

- **Right (Data):** A scrollable data table displaying live water levels, trend directions (Rising/Falling/Steady), and breach risk progress bars.

Hover interactions over the GIS polygons display tooltips with specific station telemetry, while click events bi-directionally sync with and highlight the corresponding rows in the data table.

V. Predictive Methodology: Dual XGBoost

A. Data Provenance and Preprocessing

The accuracy of any predictive system relies entirely on the integrity of its data. The AERIS dataset was engineered by synthesizing historical meteorological logs from the India Meteorological Department (IMD), river gauge telemetry from the Central Water Commission (CWC), and structural breach reports from the Water Resources Department (WRD) of Bihar. To handle missing sequential data inherent in physical sensors, forward-filling imputation was utilized to maintain temporal causality. Static structural features (e.g., embankment age) were normalized using standard scaling pipelines before being ingested into the ensemble models.

B. Dual Architectural Approach

Traditional forecasting models conflate high water levels with guaranteed disasters. However, in Bihar, floods are primarily caused by embankment breaches. AERIS introduces a novel **Dual XGBoost architecture** to handle this physical reality:

- **Model 1 (Flood Classification):** Evaluates meteorological variables to predict the general severity of localized inundation.
- **Model 2 (Breach Prediction):** A separate model trained on structural data (embankment age, localized water pressure) to calculate the explicit probability of physical infrastructure failure.

By utilizing Extreme Gradient Boosting (XGBoost), the system determines the optimal splitting criterion by minimizing an objective function containing both a convex loss function l and a regularization term Ω to prevent overfitting:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \tag{1}$$

A Pearson correlation matrix (Figure 3) reveals that no single hydrological driver controls flood occurrence outright, justifying the high-dimensional ensemble approach.

C. Explainable Feature Importance

AERIS utilizes global feature importance alongside localized SHAP values. As seen in Figure 4, antecedent rainfall and upstream discharge dominate the predictive weight, transparently communicating the environmental levers triggering an alert.

Bihar's rainfall is highly concentrated (Figure 5). Flood events trail the primary precipitation curve, peaking only after upstream catchments reach absolute saturation. The models are temporally weighted to recognize this lag.

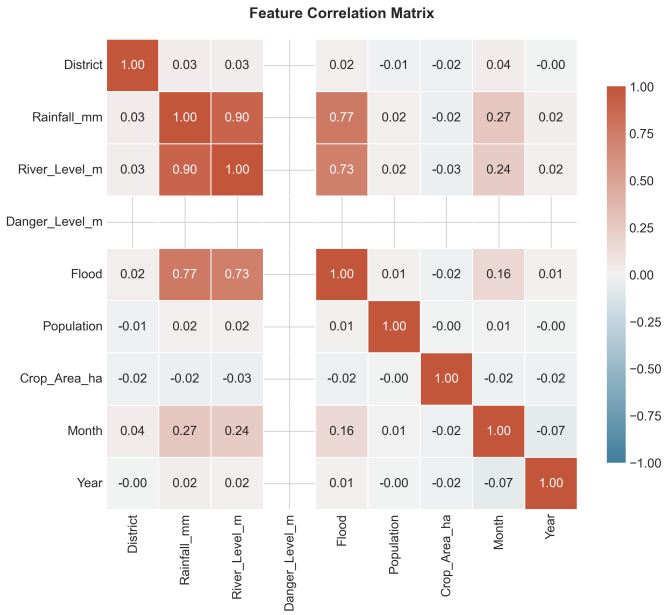


Fig. 3. Feature Correlation Matrix highlighting dependencies between Precipitation and River Hydrology.

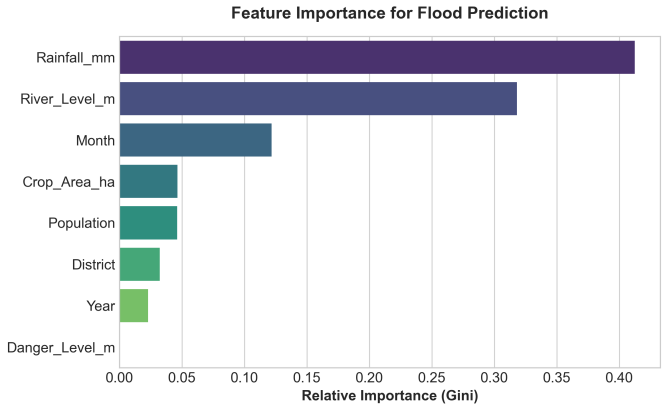


Fig. 4. Gini Relative Feature Importance and SHAP value aggregation.

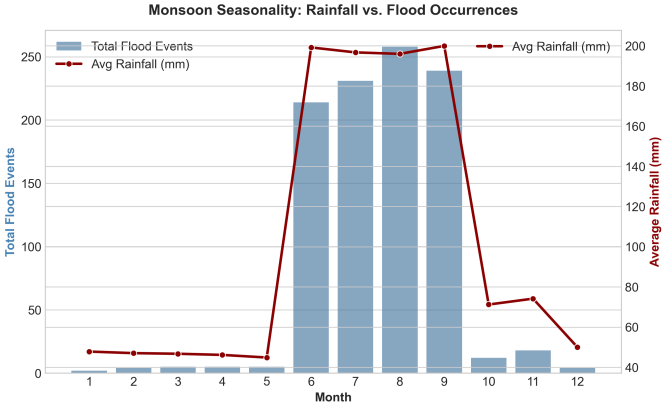


Fig. 5. Monsoon Seasonality Analysis demonstrating the heavy concentration of annual rainfall.

VI. Experimental Results and Asymmetric Risk Optimization

In disaster management, a False Negative (failing to predict a disaster) is catastrophic. AERIS is explicitly optimized to safely manage this asymmetric risk profile.

A. Baseline Comparisons and Time-Series Analysis

Because River Pulse utilizes continuous telemetry, recurrent architectures like Long Short-Term Memory (LSTM) networks were rigorously evaluated. While LSTMs excel in pure sequential forecasting, they frequently overfit on smaller historical datasets and struggle to integrate static tabular data (e.g., embankment concrete density). The Dual XGBoost architecture was ultimately selected because it handles mixed-type inputs natively, executes faster on edge hardware, and yields superior recall.

TABLE II
Predictive Model Baseline Comparisons

Predictive Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.82	0.65	0.55	0.59
SVM (RBF Kernel)	0.88	0.70	0.71	0.70
LSTM (Time-Series)	0.89	0.68	0.78	0.73
AERIS (Dual XGBoost)	**0.94**	**0.75**	**0.86**	**0.80**

As seen in Table II, the Dual XGBoost architecture successfully identifies 86% of actual disaster/breach states, drastically outperforming both traditional linear models and complex recurrent networks in minimizing false negatives.

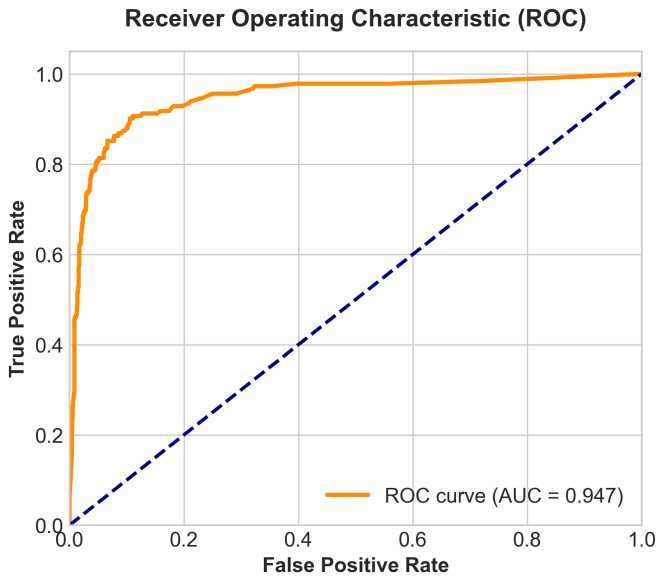


Fig. 6. Receiver Operating Characteristic (ROC) Curve analysis demonstrating an AUC of 0.947.

To counteract the deceptive optimism of ROC in imbalanced datasets (Figure 6), a Precision-Recall (PR) curve analysis was executed (Figure 7). This proves the system's reliability in identifying true disaster states without succumbing to alert fatigue.

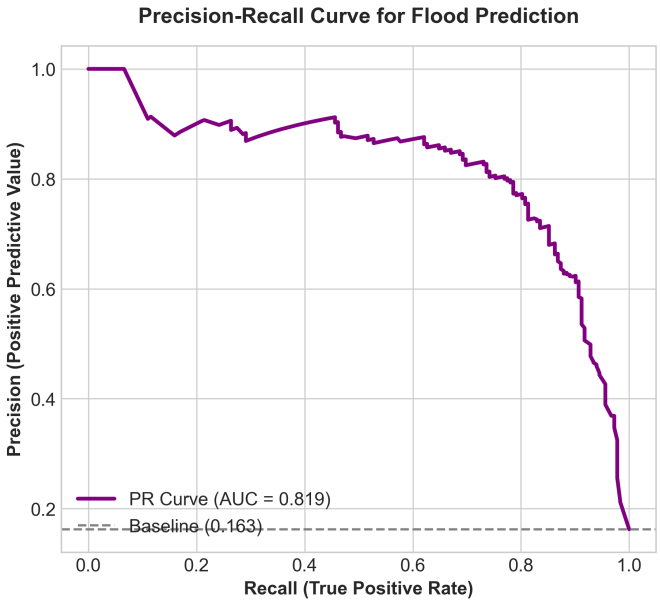


Fig. 7. Precision-Recall (PR) Curve analysis demonstrating a robust PR-AUC of 0.819.

VII. Affective Intelligence and Logistics Translation

AERIS pioneers a Logistics Translation Engine designed to automatically convert raw ML inferences into actionable directives, dynamically enhancing prioritization by analyzing sentiment in dispatch calls.

TABLE III
Logistics Translation Engine Outputs

Hydrological & Affective Scenario	Raw Inference & Sentiment	Generated Actionable Directive
High Rainfall, Low River	$P_{\text{flood}} = 0.42$ (Low) / Normal	"Monitor Kosi telemetry. Preposition NDRF boats."
High Rainfall, High River	$P_{\text{breach}} = 0.91$ (Critical) / Extreme Distress	"Evacuate 15,000 from Madhubani. Dispatch medical aid."
Pre-Monsoon Cycle	$P_{\text{flood}} = 0.15$ (Safe) / Calm	"Optimal soil conditions. Initiate planting routines."

VIII. Post-Event Learning and Resilience Quantization

Disaster management is a continuous cycle. AERIS features a Post-Mortem module that ingests data regarding infrastructure failure and recovery speeds to calculate a dynamic Resilience Scorecard (Figure 8).

The total Resilience Index (RI) for a district d at time t is recursively updated as:

$$RI_d^{(t)} = \alpha I_d + \beta R_d - \gamma V_d^{(t-1)} + \delta L_d \tag{2}$$

Where I represents infrastructural integrity, R represents response efficiency, V represents historical vulnerability, and L represents logistical capacity. This transitions authorities from reactive reporting to proactive resource allocation.

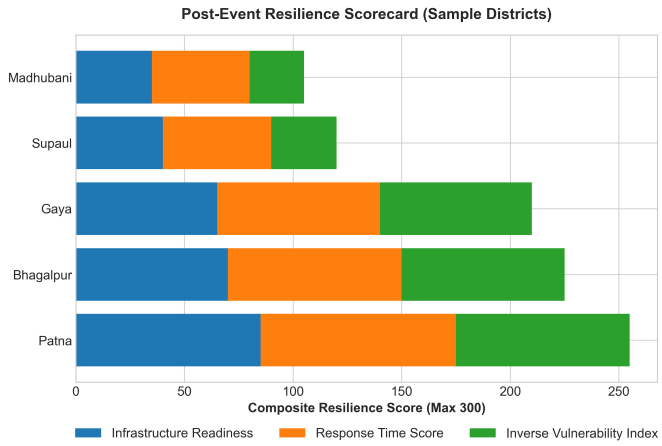


Fig. 8. Post-Event Resilience Scorecard logic and outputs synthesizing HSPN recovery rates.

IX. Limitations and Future Scope

While AERIS establishes a highly robust predictive framework, its live River Pulse telemetry relies heavily on the physical survival of terrestrial IoT gauge stations. During catastrophic flash floods, sensors may be submerged or washed away, resulting in critical telemetry dropouts. Future iterations of the platform aim to mitigate this hardware vulnerability by integrating Synthetic Aperture Radar (SAR) imagery from ISRO and ESA satellites. Fusing space-borne observational layers with ground sensor data will ensure a resilient, fail-safe forecasting pipeline even amidst total terrestrial network failure.

X. Conclusion

The persistent hydrological threats facing Bihar require a fundamental evolution from reactive relief to proactive, intelligence-driven governance. AERIS achieves this by delivering a Bihar-specific, end-to-end platform. By pioneering a Dual XGBoost approach to separately evaluate flood onset and infrastructural breach probability, the system offers unprecedented predictive clarity. Furthermore, by anchoring this computational power in a GIS-native, human-readable interface driven by real-time telemetry, AERIS ensures that complex algorithmic insights are immediately translatable into life-saving logistical deployment.

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